SQL QUERY IMPLEMENTATION

PURPLLE ONLINE BEAUTY SHOPPING

Introduction

The Purplle Online Beauty Shopping Database Management System is designed to manage product, customer, and order details efficiently. SQL queries are used to retrieve and organize data stored in the database. Commands such as SELECT, WHERE, ORDER BY, and DISTINCT help access and filter records. Conditions such as AND, OR, IN, and BETWEEN are used to retrieve specific information. Business reports help analyze product prices, customer orders, and purchase amounts.

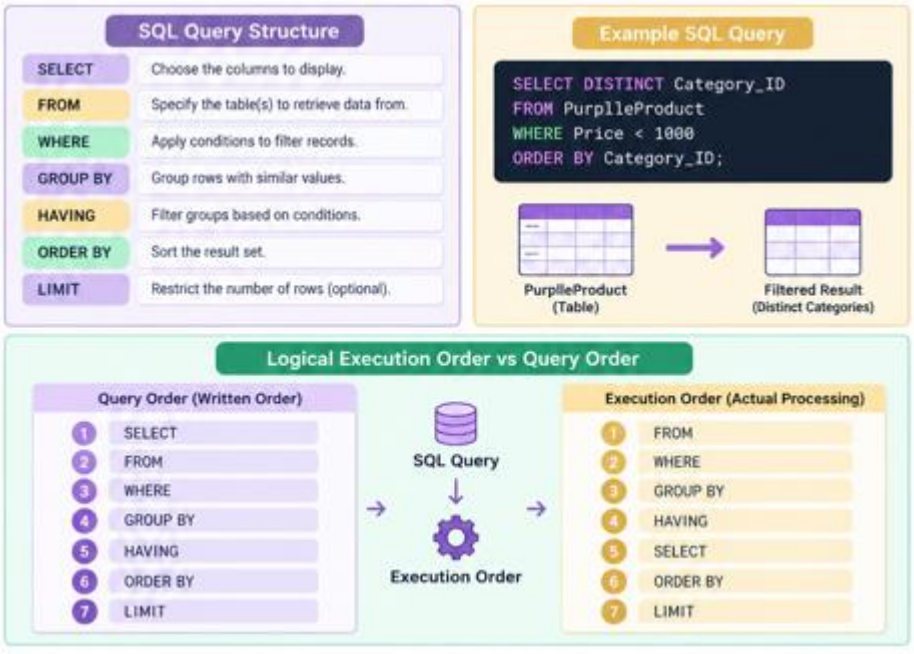
Objectives

1. Perform SELECT, WHERE, ORDER BY, and DISTINCT queries.
2. Search products based on price, category, and availability.
3. Retrieve customer and product information.
4. Apply filtering conditions.
5. Generate basic business reports.

1. Basic SQL Queries

1.1 SELECT Query

The SELECT statement is used to retrieve data from a table.



Display all products

```
SQL> SELECT *
  2  FROM PurplleProduct;

PRODUCT_ID
----------
PRODUCT_NAME
--------------------------------------------------------------------------------
CATEGORY_ID   BRAND_ID  SELLER_ID   ADMIN_ID
----------- ---------- ---------- ----------
DESCRIPTION
--------------------------------------------------------------------------------
EXPIRY_DA      PRICE      STOCK
--------- ---------- ----------
       101
Lakme Foundation
          1        201        301        401

PRODUCT_ID
----------
PRODUCT_NAME
--------------------------------------------------------------------------------
CATEGORY_ID   BRAND_ID  SELLER_ID   ADMIN_ID
----------- ---------- ---------- ----------
DESCRIPTION
--------------------------------------------------------------------------------
EXPIRY_DA      PRICE      STOCK
--------- ---------- ----------
Liquid foundation
10-MAY-27        699         50


PRODUCT_ID
----------
PRODUCT_NAME
--------------------------------------------------------------------------------
CATEGORY_ID   BRAND_ID  SELLER_ID   ADMIN_ID
----------- ---------- ---------- ----------
DESCRIPTION
--------------------------------------------------------------------------------
EXPIRY_DA      PRICE      STOCK
--------- ---------- ----------
       102
Maybelline Lipstick
          1        202        302        401

PRODUCT_ID
----------
PRODUCT_NAME
--------------------------------------------------------------------------------
CATEGORY_ID   BRAND_ID  SELLER_ID   ADMIN_ID
```

```
Fragrance perfume
10-MAR-29        599         25


PRODUCT_ID
----------
PRODUCT_NAME
--------------------------------------------------------------------------------
CATEGORY_ID   BRAND_ID  SELLER_ID   ADMIN_ID
----------- ---------- ---------- ----------
DESCRIPTION
--------------------------------------------------------------------------------
EXPIRY_DA      PRICE      STOCK
--------- ---------- ----------
       108
Nivea Body Lotion
          5        208        308        405

PRODUCT_ID
----------
PRODUCT_NAME
--------------------------------------------------------------------------------
CATEGORY_ID   BRAND_ID  SELLER_ID   ADMIN_ID
----------- ---------- ---------- ----------
DESCRIPTION
--------------------------------------------------------------------------------
EXPIRY_DA      PRICE      STOCK
--------- ---------- ----------
Moisturizing body lotion
15-APR-28        299         55


PRODUCT_ID
----------
PRODUCT_NAME
--------------------------------------------------------------------------------
CATEGORY_ID   BRAND_ID  SELLER_ID   ADMIN_ID
----------- ---------- ---------- ----------
DESCRIPTION
--------------------------------------------------------------------------------
EXPIRY_DA      PRICE      STOCK
--------- ---------- ----------
       110
Lakme Kajal
          1        201        301        401

PRODUCT_ID
----------
PRODUCT_NAME
--------------------------------------------------------------------------------
CATEGORY_ID   BRAND_ID  SELLER_ID   ADMIN_ID
----------- ---------- ---------- ----------
DESCRIPTION
--------------------------------------------------------------------------------
EXPIRY_DA      PRICE      STOCK
--------- ---------- ----------
Eye makeup kajal
10-JUL-27        299         70


10 rows selected.
```

Display selected product details

```
SQL> SELECT Product_ID, Product_Name, Price
  2  FROM PurplleProduct;

PRODUCT_ID
----------
PRODUCT_NAME
--------------------------------------------------------------------------------
     PRICE
----------
       101
Lakme Foundation
       699

       102
Maybelline Lipstick
       749

PRODUCT_ID
----------
PRODUCT_NAME
--------------------------------------------------------------------------------
     PRICE
----------

       103
Mamaearth Face Wash
       349

       104
Minimalist Serum

PRODUCT_ID
----------
PRODUCT_NAME
--------------------------------------------------------------------------------
     PRICE
----------
       699

       105
Loreal Shampoo
       399

       106

PRODUCT_ID
----------
PRODUCT_NAME
--------------------------------------------------------------------------------
     PRICE
----------
Tresemme Conditioner
       449

       109
Dove Body Wash
       349
```

1.2 WHERE Clause

The WHERE clause is used to filter records based on a condition. Find products with price less than 500

```
SQL> SELECT Product_ID, Product_Name, Price, Stock
  2  FROM PurplleProduct
  3  WHERE Price < 500;

PRODUCT_ID
----------
PRODUCT_NAME
--------------------------------------------------------------------------------
     PRICE      STOCK
---------- ----------
       103
Mamaearth Face Wash
       349         60

       105
Loreal Shampoo
       399         45

PRODUCT_ID
----------
PRODUCT_NAME
--------------------------------------------------------------------------------
     PRICE      STOCK
---------- ----------

       106
Tresemme Conditioner
       449         30

       109
Dove Body Wash

PRODUCT_ID
----------
PRODUCT_NAME
--------------------------------------------------------------------------------
     PRICE      STOCK
---------- ----------
       349         40

       108
Nivea Body Lotion
       299         55

       110

PRODUCT_ID
----------
PRODUCT_NAME
--------------------------------------------------------------------------------
     PRICE      STOCK
---------- ----------
Lakme Kajal
       299         70


6 rows selected.
```

Find products with price greater than 100

```
SQL> SELECT Product_ID, Product_Name, Price, Stock
  2  FROM PurplleProduct
  3  WHERE Price > 500;

PRODUCT_ID
----------
PRODUCT_NAME
--------------------------------------------------------------------------------
     PRICE      STOCK
---------- ----------
       101
Lakme Foundation
       699         50

       102
Maybelline Lipstick
       749         60

PRODUCT_ID
----------
PRODUCT_NAME
--------------------------------------------------------------------------------
     PRICE      STOCK
---------- ----------

       104
Minimalist Serum
       699         35

       107
Fogg Perfume

PRODUCT_ID
----------
PRODUCT_NAME
--------------------------------------------------------------------------------
     PRICE      STOCK
---------- ----------
       599         25
```

1.3 ORDER BY

The ORDER BY clause is used to sort the query results.

Products from lowest to highest price

```
SQL> SELECT Product_ID, Product_Name, Price
  2  FROM PurplleProduct
  3  ORDER BY Price ASC;

PRODUCT_ID
----------
PRODUCT_NAME
--------------------------------------------------------------------------------
     PRICE
----------
       108
Nivea Body Lotion
       299

       110
Lakme Kajal
       299

PRODUCT_ID
----------
PRODUCT_NAME
--------------------------------------------------------------------------------
     PRICE
----------

       109
Dove Body Wash
       349

       103
Mamaearth Face Wash

PRODUCT_ID
----------
PRODUCT_NAME
--------------------------------------------------------------------------------
     PRICE
----------
       349

       105
Loreal Shampoo
       399

       106

PRODUCT_ID
----------
PRODUCT_NAME
--------------------------------------------------------------------------------
     PRICE
----------
Tresemme Conditioner
       449

       107
Fogg Perfume
       599


PRODUCT_ID
----------
PRODUCT_NAME
```

Products from highest to lowest price

```
SQL> SELECT Product_ID, Product_Name, Price
  2  FROM PurpleProduct
  3  ORDER BY Price DESC;

PRODUCT_ID
----------
PRODUCT_NAME
--------------------------------------------------------------------------------
     PRICE
----------
       102
Maybelline Lipstick
       749

       101
Lakme Foundation
       699

PRODUCT_ID
----------
PRODUCT_NAME
--------------------------------------------------------------------------------
     PRICE
----------

       104
Minimalist Serum
       699

       107
Fogg Perfume

PRODUCT_ID
----------
PRODUCT_NAME
--------------------------------------------------------------------------------
     PRICE
----------
       599

       106
Tresemme Conditioner
       449

       105

PRODUCT_ID
----------
PRODUCT_NAME
--------------------------------------------------------------------------------
     PRICE
----------
Loreal Shampoo
       399

       109
Dove Body Wash
       349


PRODUCT_ID
----------
PRODUCT_NAME
```

1.4 DISTINCT

The DISTINCT keyword is used to display unique values.

Display unique categories

```
SQL> SELECT DISTINCT Product_ID, Product_Name, Price
  2  FROM PurplleProduct
  3  ;

PRODUCT_ID
----------
PRODUCT_NAME
--------------------------------------------------------------------------------
     PRICE
----------
       101
Lakme Foundation
       699

       102
Maybelline Lipstick
       749

PRODUCT_ID
----------
PRODUCT_NAME
--------------------------------------------------------------------------------
     PRICE
----------

       103
Mamaearth Face Wash
       349

       104
Minimalist Serum

PRODUCT_ID
----------
PRODUCT_NAME
--------------------------------------------------------------------------------
     PRICE
----------
       699

       105
Loreal Shampoo
       399

       106

PRODUCT_ID
----------
PRODUCT_NAME
--------------------------------------------------------------------------------
     PRICE
----------
Tresemme Conditioner
       449

       109
Dove Body Wash
```

2. Search Products

2.1 Search Products Based on Price

This displays products whose price is between ₹100 and ₹400.

```
SQL> SELECT Product_ID, Product_Name, Price
  2  FROM PurplleProduct
  3  WHERE Price BETWEEN 100 AND 400;

PRODUCT_ID
----------
PRODUCT_NAME
--------------------------------------------------------------------------------
     PRICE
----------
       103
Mamaearth Face Wash
       349

       105
Loreal Shampoo
       399

PRODUCT_ID
----------
PRODUCT_NAME
--------------------------------------------------------------------------------
     PRICE
----------

       109
Dove Body Wash
       349

       108
Nivea Body Lotion

PRODUCT_ID
----------
PRODUCT_NAME
--------------------------------------------------------------------------------
     PRICE
----------
       299

       110
Lakme Kajal
       299
```

2.2 Search Products Based on Category

This displays products belonging to category 1.

```
SQL> SELECT Product_ID, Product_Name, Category_ID, Price
  2  FROM PurplleProduct
  3  WHERE Category_ID = 1;

PRODUCT_ID
----------
PRODUCT_NAME
--------------------------------------------------------------------------------
CATEGORY_ID      PRICE
----------- ----------
       101
Lakme Foundation
          1        699

       102
Maybelline Lipstick
          1        749

PRODUCT_ID
----------
PRODUCT_NAME
--------------------------------------------------------------------------------
CATEGORY_ID      PRICE
----------- ----------

       110
Lakme Kajal
          1        299
```

3. Retrieve Customer Information

Display all customers

```
SQL> SELECT Customer_ID,First_Name, Email,Phone
  2  FROM Customer5;

CUSTOMER_ID FIRST_NAME
----------- --------------------------------------------------
EMAIL
--------------------------------------------------------------------------------
PHONE
--------------
       101 Nikitha
nikitha@gmail.com
9876543210

       102 Priya
priya@gmail.com
9876501234

CUSTOMER_ID FIRST_NAME
----------- --------------------------------------------------
EMAIL
--------------------------------------------------------------------------------
PHONE
--------------

       103 Arun
arun@gmail.com
9876512345

       104 Rahul
rahul.verma@gmail.com

CUSTOMER_ID FIRST_NAME
----------- --------------------------------------------------
EMAIL
--------------------------------------------------------------------------------
PHONE
--------------
9876523456

       105 Sneha
sneha@gmail.com
9876534567

       106 Anjali

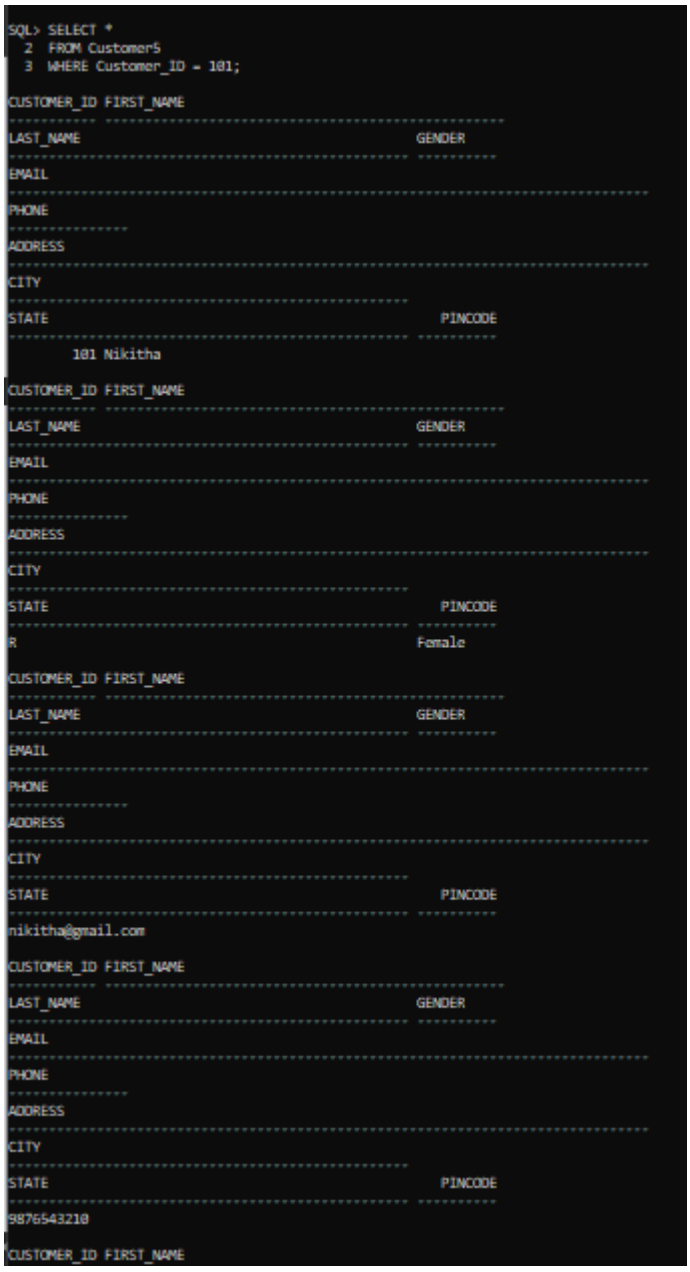
CUSTOMER_ID FIRST_NAME
----------- --------------------------------------------------
EMAIL
--------------------------------------------------------------------------------
PHONE
--------------
anjali@gmail.com
9876545678

       107 Karthik
karthik@gmail.com
9876556789


CUSTOMER_ID FIRST_NAME
----------- --------------------------------------------------
EMAIL
--------------------------------------------------------------------------------
PHONE
--------------
       108 Divya
divya@gmail.com
9876567890


8 rows selected.
```

Search for a particular customer



```
SQL> SELECT *
  2  FROM Customer5
  3  WHERE Customer_ID = 101;

CUSTOMER_ID FIRST_NAME
----------- --------------------------------------------------
LAST_NAME                                          GENDER
-------------------------------------------------- ----------
EMAIL
--------------------------------------------------------------------------------
PHONE
---------------
ADDRESS
--------------------------------------------------------------------------------
CITY
--------------------------------------------------
STATE                                                PINCODE
-------------------------------------------------- ----------
        101 Nikitha

CUSTOMER_ID FIRST_NAME
----------- --------------------------------------------------
LAST_NAME                                          GENDER
-------------------------------------------------- ----------
EMAIL
--------------------------------------------------------------------------------
PHONE
---------------
ADDRESS
--------------------------------------------------------------------------------
CITY
--------------------------------------------------
STATE                                                PINCODE
-------------------------------------------------- ----------
R                                                  Female

CUSTOMER_ID FIRST_NAME
----------- --------------------------------------------------
LAST_NAME                                          GENDER
-------------------------------------------------- ----------
EMAIL
--------------------------------------------------------------------------------
PHONE
---------------
ADDRESS
--------------------------------------------------------------------------------
CITY
--------------------------------------------------
STATE                                                PINCODE
-------------------------------------------------- ----------
nikitha@gmail.com

CUSTOMER_ID FIRST_NAME
----------- --------------------------------------------------
LAST_NAME                                          GENDER
-------------------------------------------------- ----------
EMAIL
--------------------------------------------------------------------------------
PHONE
---------------
ADDRESS
--------------------------------------------------------------------------------
CITY
--------------------------------------------------
STATE                                                PINCODE
-------------------------------------------------- ----------
9876543210

CUSTOMER_ID FIRST_NAME
```

```
9876543210

CUSTOMER_ID FIRST_NAME
----------- --------------------------------------------------
LAST_NAME                                          GENDER
-------------------------------------------------- ----------
EMAIL
--------------------------------------------------------------------------------
PHONE
---------------
ADDRESS
--------------------------------------------------------------------------------
CITY
--------------------------------------------------
STATE                                                PINCODE
-------------------------------------------------- ----------
12 Anna Nagar

CUSTOMER_ID FIRST_NAME
----------- --------------------------------------------------
LAST_NAME                                          GENDER
-------------------------------------------------- ----------
EMAIL
--------------------------------------------------------------------------------
PHONE
---------------
ADDRESS
--------------------------------------------------------------------------------
CITY
--------------------------------------------------
STATE                                                PINCODE
-------------------------------------------------- ----------
Chennai

CUSTOMER_ID FIRST_NAME
----------- --------------------------------------------------
LAST_NAME                                          GENDER
-------------------------------------------------- ----------
EMAIL
--------------------------------------------------------------------------------
PHONE
---------------
ADDRESS
--------------------------------------------------------------------------------
CITY
--------------------------------------------------
STATE                                                PINCODE
-------------------------------------------------- ----------
Tamil Nadu                                           600040

CUSTOMER_ID FIRST_NAME
----------- --------------------------------------------------
LAST_NAME                                          GENDER
-------------------------------------------------- ----------
EMAIL
--------------------------------------------------------------------------------
PHONE
---------------
ADDRESS
--------------------------------------------------------------------------------
CITY
--------------------------------------------------
STATE                                                PINCODE
-------------------------------------------------- ----------
```

4. Apply Filtering Conditions

Multiple conditions can be combined using AND, OR, and NOT.

4.1 AND Condition

This displays products that satisfy both conditions.

```
SQL> SELECT Product_ID, Product_Name, Price, Category_ID
  2  FROM PurplleProduct
  3  WHERE price < 500
  4  AND Category_ID = 2;

PRODUCT_ID
----------
PRODUCT_NAME
--------------------------------------------------------------------------------
     PRICE CATEGORY_ID
---------- -----------
       103
Mamaearth Face Wash
       349           2
```

4.2 OR Condition

This displays products belonging to either category 101 or 102.

```
SQL> SELECT Product_ID, Product_Name, Price
  2  FROM PurplleProduct
  3  WHERE Category_ID = 1
  4  OR Category_ID = 2;

PRODUCT_ID
----------
PRODUCT_NAME
--------------------------------------------------------------------------------
     PRICE
----------
       101
Lakme Foundation
       699

       102
Maybelline Lipstick
       749

PRODUCT_ID
----------
PRODUCT_NAME
--------------------------------------------------------------------------------
     PRICE
----------

       103
Mamaearth Face Wash
       349

       104
Minimalist Serum

PRODUCT_ID
----------
PRODUCT_NAME
--------------------------------------------------------------------------------
     PRICE
----------
       699

       110
Lakme Kajal
       299
```

4.3 IN Operator

```
SQL> SELECT Product_ID, Product_Name, Price
  2  FROM PurplleProduct
  3  WHERE Category_ID IN (1,2,3);

PRODUCT_ID
----------
PRODUCT_NAME
--------------------------------------------------------------------------------
     PRICE
----------
       101
Lakme Foundation
       699

       102
Maybelline Lipstick
       749

PRODUCT_ID
----------
PRODUCT_NAME
--------------------------------------------------------------------------------
     PRICE
----------

       103
Mamaearth Face Wash
       349

       104
Minimalist Serum

PRODUCT_ID
----------
PRODUCT_NAME
--------------------------------------------------------------------------------
     PRICE
----------
       699

       105
Loreal Shampoo
       399

       106

PRODUCT_ID
----------
PRODUCT_NAME
--------------------------------------------------------------------------------
     PRICE
----------
Tresemme Conditioner
       449

       110
Lakme Kajal
       299


7 rows selected.
```

4.4 BETWEEN Operator

```
SQL> SELECT Product_ID, Product_Name, Price
  2  FROM PurppleProduct
  3  WHERE Price BETWEEN 100 AND 300;

PRODUCT_ID
----------
PRODUCT_NAME
--------------------------------------------------------------------------------
     PRICE
----------
       108
Nivea Body Lotion
       299

       110
Lakme Kajal
       299

PRODUCT_ID
----------
PRODUCT_NAME
--------------------------------------------------------------------------------
     PRICE
----------
```

5. Basic Business Reports

5.1 Product Price Report

```
SQL> SELECT
  2      Product_ID,
  3      Product_Name,
  4      Price
  5  FROM purppleProduct
  6  ORDER BY Price DESC;

PRODUCT_ID
----------
PRODUCT_NAME
--------------------------------------------------------------------------------
     PRICE
----------
       102
Maybelline Lipstick
       749

       101
Lakme Foundation
       699

PRODUCT_ID
----------
PRODUCT_NAME
--------------------------------------------------------------------------------
     PRICE
----------

       104
Minimalist Serum
       699

       107
Fogg Perfume

PRODUCT_ID
----------
PRODUCT_NAME
--------------------------------------------------------------------------------
     PRICE
----------
       599

       106
Tresemme Conditioner
       449

       105

PRODUCT_ID
----------
PRODUCT_NAME
--------------------------------------------------------------------------------
     PRICE
----------
Loreal Shampoo
       399

       109
Dove Body Wash
       349


PRODUCT_ID
----------
PRODUCT_NAME
--------------------------------------------------------------------------------
     PRICE
----------
       103
Mamaearth Face Wash
       349
```

5.2 Customer Order Report

```
SQL> SELECT
  2      c.Customer_ID,
  3      c.First_Name,
  4      o.Order_ID,
  5      o.Order_Date,
  6      o.Total_Amount
  7  FROM Customer5 c
  8  JOIN Orders o
  9      ON c.Customer_ID = o.Customer_ID
 10  ORDER BY o.Order_Date DESC;

CUSTOMER_ID FIRST_NAME                                         ORDER_ID
----------- -------------------------------------------------- ----------
ORDER_DAT TOTAL_AMOUNT
--------- ------------
        105 Sneha                                                     505
26-SEP-26         1599

        104 Rahul                                                     504
26-SEP-26          799

        103 Arun                                                      503
26-SEP-26         1299


CUSTOMER_ID FIRST_NAME                                         ORDER_ID
----------- -------------------------------------------------- ----------
ORDER_DAT TOTAL_AMOUNT
--------- ------------
        102 Priya                                                     502
26-SEP-26          749

        101 Nikitha                                                   501
20-SEP-26         1699
```

5.3 Product Order Report

```
SQL> SELECT
  2      od.Order_ID,
  3      od.Product_ID,
  4      od.Quantity,
  5      od.Unit_Price,
  6      (od.Quantity * od.Unit_Price) AS Subtotal
  7  FROM Order_Details od
  8  ORDER BY Subtotal DESC;

  ORDER_ID PRODUCT_ID   QUANTITY UNIT_PRICE   SUBTOTAL
---------- ---------- ---------- ---------- ----------
       501        101          3        799       2397
       505        105          2      799.5       1599
       503        103          2      649.5       1299
       504        104          1        799        799
       502        102          1        749        749
```

5.4 Customer-wise Total Orders

```
SQL> SELECT
  2      c.Customer_ID,
  3      c.First_Name || ' ' || c.Last_Name AS Customer_Name,
  4      COUNT(o.Order_ID) AS Total_Orders
  5  FROM Customer5 c
  6  JOIN Orders o
  7      ON c.Customer_ID = o.Customer_ID
  8  GROUP BY c.Customer_ID, c.First_Name, c.Last_Name
  9  ORDER BY Total_Orders DESC;

CUSTOMER_ID
-----------
CUSTOMER_NAME
--------------------------------------------------------------------------------
TOTAL_ORDERS
------------
       101
Nikitha R
           1

       102
Priya Sharma
           1

CUSTOMER_ID
-----------
CUSTOMER_NAME
--------------------------------------------------------------------------------
TOTAL_ORDERS
------------

       105
Sneha Patel
           1

       104
Rahul Verma

CUSTOMER_ID
-----------
CUSTOMER_NAME
--------------------------------------------------------------------------------
TOTAL_ORDERS
------------
           1

       103
Arun Kumar
           1
```

5.5 Customer-wise Total Purchase Amount

```
SQL> SELECT
  2      c.Customer_ID,
  3      c.First_Name,
  4      SUM(o.Total_Amount) AS Total_Purchase
  5  FROM Customer5 c
  6  JOIN Orders o
  7      ON c.Customer_ID = o.Customer_ID
  8  GROUP BY c.Customer_ID, c.First_Name
  9  ORDER BY Total_Purchase DESC;

CUSTOMER_ID FIRST_NAME                                         TOTAL_PURCHASE
----------- -------------------------------------------------- --------------
       101 Nikitha                                                      1699
       105 Sneha                                                        1599
       103 Arun                                                         1299
       104 Rahul                                                         799
       102 Priya                                                         749
```

Conclusion

The SQL Query Implementation task was completed successfully using the Purplle Online Beauty Shopping Database Management System. Various SQL commands were used to retrieve, filter, and sort database records. Customer and product information was accessed using appropriate queries. Aggregate functions such as COUNT() and SUM() were used to generate business reports. This task improved understanding of SQL queries and database management.